Pre-feasibility study to establish a marble factory in Ajloun

2017

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1. Exclusive summary

Based on the large urban expansion in the buildings construction in Jordan, the marble ore in Ajloun area has an economic value to extract and produce panels to be used in various construction purposes, where Ajloun Marble is characterized by hardness high, low porosity and different colours. so, the idea of extracting and processing marble is economically feasible.

Based on the proposed market share and prices, which have been considered for the proposed project, the estimated revenues will be 3,211,168 Jordanian dinars during the first year of the life of the project and gradually rising to 4,189,846 Jordanian dinars during the tenth year, the payback period is 3 years and the internal rate of return is 35.0%. In addition of creating 13 jobs oppotunities.

Table 1 Project Main Indicators

Indicator	
Total Investment Cost (Jordan Dinars)	1,468,057
Internal Rate of Return (IRR)	%35.0
Pay-Back Period (Years)	3
Net Present Value (Jordan Dinars)	3,210,953
Job Opportunities provided by the Project	13
Total Revenues (For 10 Years)	36,812,442
Total Operating Expenses (For 10 years)	29,762,860
Total Net Profit (For 10 years)	2,908,296

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

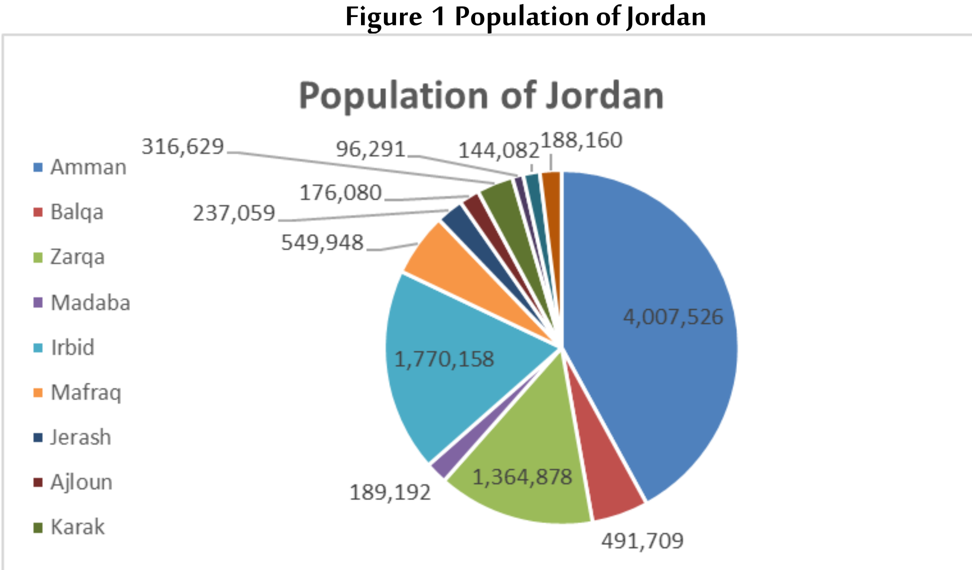
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**
15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

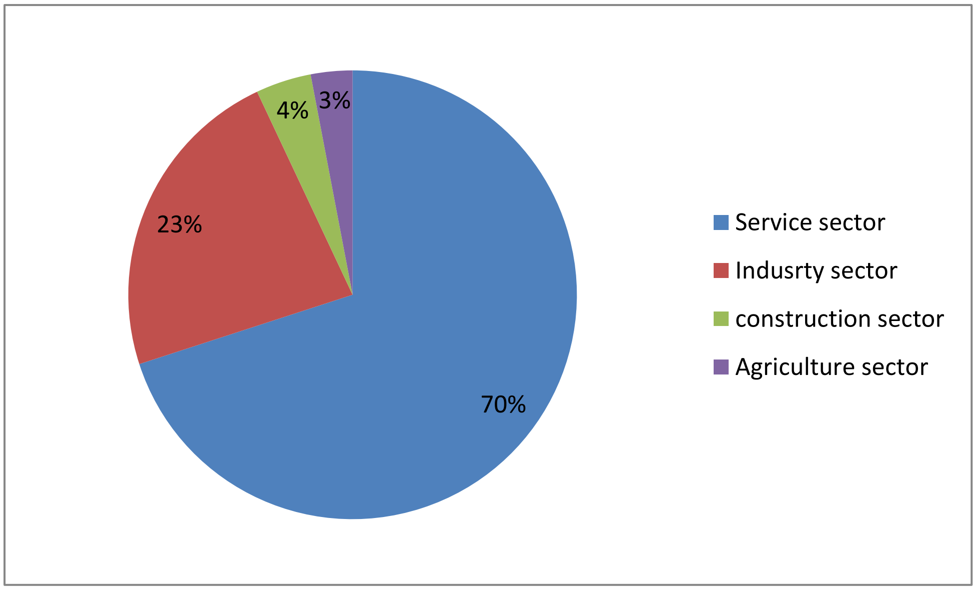
Department of Statistics (DOS), 2015.

Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.
Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.
Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy s lowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.
The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

2. Market Analysis

Project Description

Based on the local reserve / Jordanian ores of marble which is about 1.415 million m^3, the idea of the exploitation of marble ore that starts by extracting, and end use in decorative purposes, decoration and construction works can be considered as feasible idea which has benefits to the local community in terms of creating jobs and providing the local market of marble products with local competitive advantages and increasing competitiveness to up grade the quality of the production of marble, as well as it well project be economically feasible for the investor.

Marble

Calcline metamorphic rock consists of very pure calcite (crystalline form of calcium carbonate). Used in sculpture, as well as used as constructive material, and in many other purposes. It have been formulated under rare conditions of pressure and temperature underground. Most famous several country in are well known in production of marble as: Palestine, Turkey, Italy, which is the most famous country.

The marble is divided into two types:

• Natural marble: two sections the first was originally sedimentary or metamorphic rocks, and the other is originally Igneous rocks

.

• industrial marble: as a result of a chemical manufacturing which is based on certain steps to get finally a material which can be used serve as a substitute of natural marble in the construction works.

Marble is also classified according to its chemical composition to:

• Alcalceta marble mainly consists of calcite material, and five percent of the material is magnesium carbonate.

• Dolomitic Marble: In this type of marble the contents of the material of magnesium carbonates is increasing to be ranging between (5-45%) of the total composition, so it is founded in several gradations of colours, based on the proportion of the presence of this material in it.

• Alakiqi Marble: This type of marble is characterized and pure colour; and because of its contents of calcium carbonate, which was formed slowly due to deposition of cold water springs. The Alakiqi marble is spreads in yellow, brown, greencolour and distinguished mainly by transparent layers formation, which increases the beauty of its shape, and its rarity.

• Marble vivid stone: This type of marble contain varying proportions of natural impurities, which makes it a mixed chromatically, between roasy and gray colour, and it contains abundant levels of metal serpentine " hydro magnesium silica" that increases the hardness, the unit colours.

• Marble travertine: This type of marble is a chemical deposition material calcite dissolved in hot spring water, which is usually saturated with carbon dioxide, leading to be formed as monolithic layers, containing crystalline pores, varying in size

The following table shows the marble acurances in Jordan:

Table 3 The marble ocurances in Jordan

Area	Reserve (m^3)
West dabaa	60,000
Eest dabaa	130,000
West tlol alhamam	120,000
Tlol alhamam	260,000
Alkerb alromanya	60,000
South swaqa	560,000
Mqala alasad	225,000
Ajloun area	200,000
total	**1,615,000**

Source: Energy and Mineral regularity Commision

Project Objectives

The project objectives are summarized in the following points:

• The project aims to extract and produce some of industrial forms of marble which complyes with the requirements of the domestic and market export which can be used either indoors or outside in order to achieve diversity in the proposed production of this ore.

• Due to the high unemployment rate in Ajloun, the project will create jobs opportunities for the local comunity.

• Increasing the export of raw marble and open prospects for export.

• The demand of marble increased recent ly because of the massive urban revolution as one of its important components either locally or extremaly, and that allows to open the prospects for export and achieves one of the goals of the project.

Proposed Product

Marble have many uses, including:

I. used in the construction of internal and external works:

1. for the floors and stairs: Marble gives a beautiful sight calm and grandeur of the place, but should not be used on floors that are exposed to tremendous pressure and absortion because of high absorption rate of marble were the granite absorption rate is very low.

2. The external faces of buildings: It is possible to use marble in the external faces of the buildings but the texture is different from the smooth texture applied to the floor. A coarse marble texture will be used. Weight and thickness and marble area should be taken into account during the planning.

3. Walls: The same marble used for external faces, the only difference lies in the texture, as the texture of the walls will be smooth.

4. Columns: The marble used in columns decoration to give a modern luxury tio its appearance.

5. Bathrooms, laundries: There are promising colours and shapes to decorate bathrooms and laundries. For sinks it should be pre-selected marble in order to cut to fit the sintes, the thickness that are often selected for this purpose is 3 centimeters.

6. Kitchen: Preferably use dark colours for the counter top and add insulating and polishing, materials. As for the thickness it is recommended to choose 3 cm.

I. Other uses:

2. Covering the area adjacent to the door frames and windows.

3. Design sculptures and art pieces.

Targeted categories

The proposed project targeting the following categories:

• luxury company for marble and granite.

• ACCESS Company for Natural Stone

• Kairouan commercial company.

• Other companies.

Market Size Analysis

Marble domestic production amounted to about 273 340 m3 and the volume of exports reached 9340 m^3 Based on these statistics referred to the statistics of the Ministry of Energy and mineral Resources of Jordan, the establishment of a project for the extraction and processing of Jordanian marble ores in Ajloun area, which is characterized by high quality will encourages investors to be exploited. The approvals should be taken for the necessary energy and minerals Regulatory Commision.

The size of marble market depends on the size of the construction workes in Jordan. The following table shows the construction activity in Jordan, based on construction licenses and construction area (area of a thousand meters) during the period (2012-2016):

Table 4 Volume of construction in the Kingdom during the period (2012-2016), (1000 m^2)

Period	2012	2013	2014	2015	2016
Housing					
Licenses Number	**29,940**	**33,906**	**36,642**	**32,882**	**36,367**
Amman	7,956	8,096	8,904	7,596	7,172
Irbid	8,429	10,063	11,118	10,496	11,293
Zarqa	2,804	3,223	3,103	2,765	3,292
Others	10,751	12,524	13,517	12,025	14,610
Area (1000 m^2)	**10,691.2**	**11,817.6**	**12,578.4**	**10,879.0**	**10,799.7**
Amman	6,013.2	6,075.6	6,084.1	4,967.3	4,019.7
Irbid	1,806.0	2,243.1	2,611.9	2,378.1	2,599.3
Zarqa	756.3	942.9	983.4	888.8	1,076.3
Others	2,115.7	2,556.0	2,899.0	2,644.8	3,104.3
Other purposes					
Licenses Number	**2,526**	**2,695**	**2,936**	**2,893**	**3,043**
Amman	1,188	1,176	1,334	1,322	1,373
Irbid	431	568	672	657	547
Zarqa	338	423	471	477	600
Other s	569	528	459	437	523
Area (1000 m^2)	**2,215.5**	**2,167.5**	**2,413.9**	**2,243.8**	**2,510.5**
Amman	1,604.0	1,338.3	1,616.8	1,457.0	1,618.5
Irbid	131.1	254.8	233.6	268.9	188.6
Zarqa	146.6	175.6	265.2	217.3	294.4
Others	333.8	398.8	298.3	300.6	409.0

Source: Central Bank/ Monthly Statistical Bulletin (2016)

Competitor Analysis

Marble is one of high-exploitation of materials in Jordan, based on the local production capacity, there are many companies exploiting marble such as:

- Majid Rajabi & Bros. Co.
- • United Brothers Inc.
- • Other companies

Market Share

A quarry producing of 8,000 m^3 will be established with a market share estimated at 3% based on technical experience, its possible to increase the production capacity wich leads to increase the market share based on supply and demand in the local market.

Price Analysis and Pricing Policy

Marble existing prices used in the construction works and decoration in the domestic market is ranging between 12-17 Jordanian dinars per square meter, based on the product category, as well as, the export and import volumes of raw marble. There is a great chance to compete with production capacity estimated at 8000 m^3 / year in addition to export, the following table shows the expected annual revenues of the project.

Table 5 Expected Prices

Product	Ratio (%)	The annual production (m^3)	Annual production (M^2)	Price m^2 (JD)	Price m^3 (JD)
Main product (polished	80	6,400	213,120	15	-----
By product (created marble)	20	1,600	----	-----	7
total	**100**	**8,000**		-----	-----

It was assumed that one cubic meter of marble gives 33.33 m^2, and t the production capacity of 8,000 m^3 per year during the year (2018-2027) 3% annual price incremental have been assumed as well.

Expected Revenues Analysis

Table6 Projected annual revenues for 10 years (2027-2018)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price of the meter square of polished marble	15.00	15.45	15.91	16.39	16.88	17.39	17.91	18.45	19.00	19.57
Price of cubic meter of cracked marble.	7.00	7.21	7.43	7.65	7.88	8.11	8.36	8.61	8.87	9.13
Polished marble revenues	3199968	3295967	3394846	3496691	3601592	3709640	3820929	3935557	4053624	4175232
cracked marble revenues	11200	11536	11882	12239	12606	12984	13373	13775	14188	14613
Total of Revenues	**3211168**	**3307503**	**3406728**	**3508930**	**3614198**	**3722624**	**3834303**	**3949332**	**4067812**	**4189846**

3. Technical analysis

Human resorces required

Table7 Human resources requird for the project

Job description	No.
Project manager	1
Technical cutter	2
Accountant	1
Labour	2
Driver	4
Security guard	2
Cleaning labour	1
total	**13**

The following is a clarification of the job description of the workforce:

• **General Manager**: proposing the general policy of the plant, preparation of plans and follow up the implementation, technical and administrative supervision of staff, development of procedures, ensuring of occupational health and safety requirements.

• **Accountant**: Follow up money aspects, financial, auditing, and preparing of the annual report of the financial situation.

• **Driver**: transporting and distribution of products to customers.

• **Cleaner**: maintaining the cleaning of the working area.

• **Guard**: guarding the site and maintaining the security of at the site.

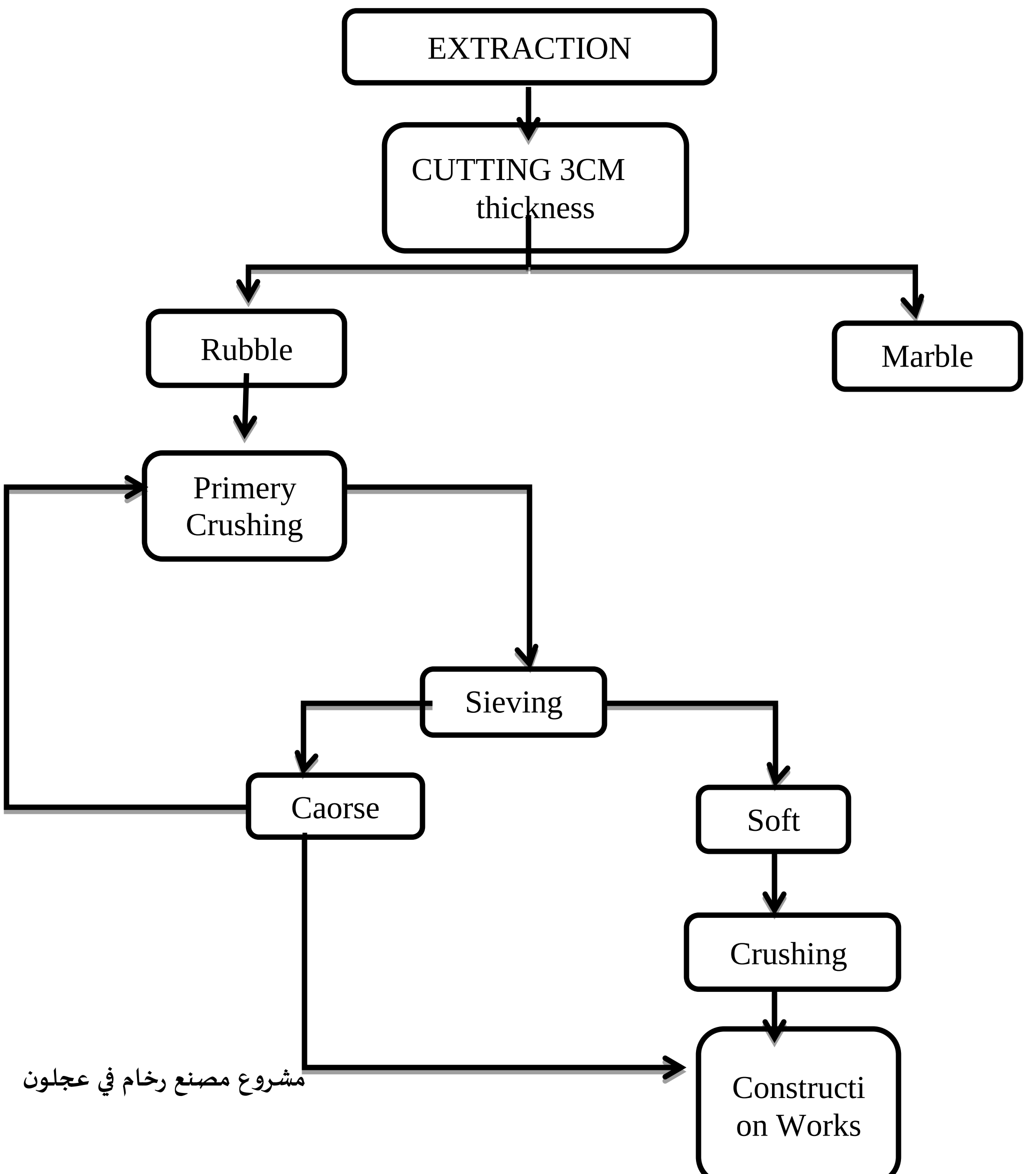
Location analysis

The plant will be the established at Ajloun area, after getting the necessary Licences from the concerned governorate and departments.

Description of production process and steps of manufacturing

Extracting marble ore from its occurances locations will be carried out by using special stone shearing to be produced in panels form to be used in the construction and decoration works, after shearing process, about 20% of the ore will be not suitable for the use of decoration and adornment (rubbles), which will be extracted by mechanical processing using a jaw crusher and then sieved to define the particles distribution; the soft will be used directly in the construction works and the rough part is divided into two parts, one returns back to the crushing stage and the other is used in construction works.

Figurre (3) Flow sheet of marble processes



Technical requirement analysis

Land and construction works

A license for quarry for agricultural lands reclamation will be taken since it is not allowed take quary license in Ajloun dut to its privacy in the large number of trees therefore its allowed to take a license for the reclamation of land and can be renewed annually and extended. According to the recommendations of the Central Committee of State Property / Energy and Minerals sector Regulatory Commission.

The construction works required for the project is shown in table (7):

Table8 construction work required

construction works	Area(M^2)
Building purposes	150
Hunger	500
Other	-

Equipments and machines

The following table shows the machinery and equipment needed for the project,:

Table 9 machinery and equipment

Machine	Number
loader	1
Crushing machine	1
Cutter machine	1
dumber	1
Seivig machine	2
Secondry crushing machine	1
small saw	2

Furniture

Table10 furnitures

Description	Cost
feurnitures	10,000

Vehicles and transportation

The following table shows a list of vehicles and transport equipment required for the project:

Table11 vehicles and transport media required for the project

Vehicles and transportation	Number
Transporte and Vehicles	2

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is %10.75 as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table12 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table13 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table14 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1)) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table15 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	65.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	2.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table16 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0.5
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	**Value**
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	10.12%
Income Tax Rate	5.00%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table17 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	587,223
Loans Value	880,834

Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	880,834
Loans Percentage	%60
Equity	
Cost of Equity	%15.5
Equity Value	587,223
Equety Percentage	%40
Gross	
Project Value	1,468,057
Weighted Average Cost of Capital	%10.75

Capital Expenditures

The estimated cost of the project is 1,468,057 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table18 Construction Work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Construction Works	200		30,000
Hungers works	100		50,000
Other Works			8,000
Total			**88,000**

Table19 machinery and equipments Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
loder	1	60,000	60,000
Crushing machine	1	40,000	40,000
Cutter machine	1	70,000	70,000
dumber	1	200,000	200,000
Seivig machine	2	25,000	50,000
Secondry crushing machine	1	40,000	40,000
Miny saw	2	25,000	50,000
Total			**510,000**

Table20 Furniture and furnishings Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
furniture	-	-	10,000
Total			**10,000**

Table21 vehicles and trasnportations Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Medium vehicle (pickup)	2	23,000	46,000
Total			**46,000**

Table22 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	**2,500**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table23 Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	649,057
General and administrative expenses	3	%25	4,193
Marketing expenses	3	%25	2,007
Indirect Salaries	3	%25	22,840
Total			**678,097**

Table24 Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Construction works	88,000	10%	96,800	%6.6
Machinery &Equipment	510,000	10%	561,000	%38.2
Furniture	10,000	10%	11,000	%0.7
Vehicles	46,000	10%	50,600	%3.4
Pre-Operating Expenses	2,500	10%	2,750	%0.2
Working capital	678,097	10%	745,907	%50.8
Total (JD)	**1,334,597**		**1,468,057**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,468,057) JD, where 60% of the project will be financed through loans (880,834) JD, and the remaining (40%) through (Equity) with (587,223) JD. The following table summarizes the general structure of the required financing:

Table25 Sources of Funding

Funding Sources	Amount	Percentage
Equity	587,223	%40
Loans	880,834	%60
Total	**1,468,057**	**%100**
Use of Fund:	**Value**	**Ratio**
Capital Expenditures	719,400	%49.00
Pre-Operating Expenses	2,750	%0.19
Working Capital	745,907	%50.81
Total	**1,468,057**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 26 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	2,087,259	2,149,877	2,214,373	2,280,804	2,349,229	2,419,705	2,492,297	2,567,066	2,644,078	2,723,400
Utilities from Revenues	160,558	165,375	170,336	175,446	180,710	186,131	191,715	197,467	203,391	209,492
Maintenance from Revenues	80,279	82,688	85,168	87,723	90,355	93,066	95,858	98,733	101,695	104,746
Disposables from Revenues	32,112	33,075	34,067	35,089	36,142	37,226	38,343	39,493	40,678	41,898
Others	236,021	243,101	250,395	257,906	265,644	273,613	281,821	290,276	298,984	307,954
Total	**2,596,229**	**2,674,116**	**2,754,340**	**2,836,970**	**2,922,079**	**3,009,741**	**3,100,034**	**3,193,035**	**3,288,826**	**3,387,490**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 27 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Staionary and Printing	600	618	637	656	675	696	716	738	760	783
Transportaton	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rent	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance	5,396	5,557	5,724	5,896	6,073	6,255	6,443	6,636	6,835	7,040
Miscellaneous	1,525	1,570	1,617	1,666	1,716	1,767	1,820	1,875	1,931	1,989
Total	**16,770**	**17,273**	**17,791**	**18,325**	**18,875**	**19,441**	**20,024**	**20,625**	**21,244**	**21,881**

Marketing Expenses

Table28 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	8,028	8,269	8,517	8,772	9,035	9,307	9,586	9,873	10,170	10,475

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table29 The projected staff number

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	1	1	1	1	1	1	1	1	1	1
Technical cutter	2	2	2	2	2	2	2	2	2	2
accounter	1	1	1	1	1	1	1	1	1	1
labour	2	2	2	2	2	2	2	2	2	2
driver	4	4	4	4	4	4	4	4	4	4
Security guard	2	2	2	2	2	2	2	2	2	2
Cleaning man	1	1	1	1	1	1	1	1	1	1
Total	**13**	**13**	**13**	**13**	**13**	**13**	**13**	**13**	**13**	**13**

Table 30 Expected monthly salary and wages for the years (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Technical cutter	380	396	412	429	446	465	484	503	524	546
accounter	400	416	433	451	470	489	509	530	552	574
labour	300	312	325	338	352	367	382	397	414	431
driver	400	416	433	451	470	489	509	530	552	574
Security guard	300	312	325	338	352	367	382	397	414	431
Cleaning man	250	260	271	282	294	306	318	331	345	359

Table 31 Expected Annual Salaries and Wges (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Technical cutter	12,160	12,659	13,178	13,718	14,280	14,866	15,475	16,110	16,770	17,458
accounter	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
labour	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
driver	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Security guard	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Cleaning man	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Total	**91,360**	**95,106**	**99,005**	**103,064**	**107,290**	**111,689**	**116,268**	**121,035**	**125,997**	**131,163**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table32 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	96,800	%5.0	%0.0
Machinaries	561,000	%10.0	%2.0
Furniture	11,000	%10.0	%5.0
Vehicles	50,600	%10.0	%0.0

Table33 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	719,400	731,170	743,192	755,472	768,015	780,828	793,918	807,291	820,953	834,912
Total Depreciation Expenses	67,100	68,277	69,479	70,707	71,962	73,243	74,552	75,889	77,255	78,651
Total Accumulated Depreciation	67,100	135,377	204,856	275,563	347,525	420,768	495,320	571,209	648,464	727,115
Total Additions	0	11,770	12,022	12,280	12,543	12,813	13,090	13,373	13,662	13,959
Total Net Book Values	652,300	595,793	538,336	479,908	420,490	360,061	298,599	236,082	172,489	107,797

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 34 Loan Details

Loan Value	880,834
Annual Interst Rate	9.48%
Loans Period	5
Loan Starts at year:	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				880,834
2019	0	83,503	-83,503	964,337
2020	0	91,419	-91,419	1,055,756
2021	0	100,086	-100,086	1,155,842
2022	0	109,574	-109,574	1,265,416
2023	0	119,961	-119,961	1,385,377
2024	0	131,334	-131,334	1,516,711
2025	0	143,784	-143,784	1,660,495
2026	0	157,415	-157,415	1,817,910
2027	0	172,338	-172,338	1,990,248
2028	0	188,675	-188,675	2,178,923

Income Statement

Table 35 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	3,211,168	3,307,503	3,406,728	3,508,930	3,614,198	3,722,624	3,834,303	3,949,332	4,067,812	4,189,846
Gross Operating Revenues	3,211,168	3,307,503	3,406,728	3,508,930	3,614,198	3,722,624	3,834,303	3,949,332	4,067,812	4,189,846
Operating Expenses	(2,596,229)	(2,674,116)	(2,754,340)	(2,836,970)	(2,922,079)	(3,009,741)	(3,100,034)	(3,193,035)	(3,288,826)	(3,387,490)
Gross Operating Profit	614,939	633,387	652,388	671,960	692,119	712,882	734,269	756,297	778,986	802,355
Gross Profit Percentage	*%19*	*%19*	*%19*	*%19*	*%19*	*%19*	*%19*	*%19*	*%19*	*%19*
Salaries and Benefits (Indirect Staff)	(91,360)	(95,106)	(99,005)	(103,064)	(107,290)	(111,689)	(116,268)	(121,035)	(125,997)	(131,163)
General and Administraive Expenses	(16,770)	(17,273)	(17,791)	(18,325)	(18,875)	(19,441)	(20,024)	(20,625)	(21,244)	(21,881)
Markeing Expenses	(8,028)	(8,269)	(8,517)	(8,772)	(9,035)	(9,307)	(9,586)	(9,873)	(10,170)	(10,475)
Pre Operating Expenses	(2,750)									
Gross Indirect Expenses	(118,908)	(120,648)	(125,313)	(130,162)	(135,200)	(140,436)	(145,878)	(151,533)	(157,411)	(163,519)
Income before Interest, Depreciation and Tax	496,031	512,739	527,075	541,798	556,919	572,446	588,391	604,764	621,575	638,836
Fixed Assets Depreciations	(67,100)	(68,277)	(69,479)	(70,707)	(71,962)	(73,243)	(74,552)	(75,889)	(77,255)	(78,651)
Income before Tax and Interests	428,931	444,462	457,596	471,091	484,957	499,203	513,839	528,874	544,320	560,185

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Bank Interests	(83,503)	(91,419)	(100,086)	(109,574)	(119,961)	(131,334)	(143,784)	(157,415)	(172,338)	(188,675)
Income Before Tax	345,428	353,043	357,510	361,517	364,996	367,869	370,055	371,460	371,982	371,510
Income Tax	(69,086)	(70,609)	(71,502)	(72,303)	(72,999)	(73,574)	(74,011)	(74,292)	(74,396)	(74,302)
Net Profit	276,342	282,434	286,008	289,214	291,997	294,296	296,044	297,168	297,586	297,208
Net Profit Percentage	*%9*	*%9*	*%8*	*%8*	*%8*	*%8*	*%8*	*%8*	*%7*	*%7*
Compulsory Reserves	(27,634)	(28,243)	(28,601)	(28,921)	(29,200)	(29,430)	(29,604)	(29,717)	(29,759)	(29,721)
Retained Earnings	248,708	502,899	760,306	1,020,599	1,283,396	1,548,262	1,814,701	2,082,152	2,349,979	2,617,466

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table36 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	276,342	282,434	286,008	289,214	291,997	294,296	296,044	297,168	297,586	297,208
Bank Interests	83,503	91,419	100,086	109,574	119,961	131,334	143,784	157,415	172,338	188,675
Depreciation	67,100	68,277	69,479	70,707	71,962	73,243	74,552	75,889	77,255	78,651
Total Operating Cashflows before Additions to Working Capital	426,945	442,131	455,573	469,495	483,919	498,872	514,380	530,472	547,179	564,534
Inventory (Increase/Decrease)	(173,938)	(5,218)	(5,375)	(5,536)	(5,702)	(5,873)	(6,049)	(6,231)	(6,418)	(6,610)
Accounts Receviables (Increase/Decrease)	84,828	2,545	2,621	2,700	2,781	2,864	2,950	3,039	3,130	3,224
Working Capital (Increase/Decrease)	(89,110)	(2,673)	(2,753)	(2,836)	(2,921)	(3,009)	(3,099)	(3,192)	(3,288)	(3,386)
Net Cashflows from Operating Activities	337,835	439,457	452,820	466,659	480,998	495,863	511,281	527,280	543,891	561,148
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(719,400)	(11,770)	(12,022)	(12,280)	(12,543)	(12,813)	(13,090)	(13,373)	(13,662)	(13,959)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Cashflows from Investments	(719,400)	(11,770)	(12,022)	(12,280)	(12,543)	(12,813)	(13,090)	(13,373)	(13,662)	(13,959)
Cashflows from Financing Activities										
Capital	587,223									
Return the loan	83,503	91,419	100,086	109,574	119,961	131,334	143,784	157,415	172,338	188,675
Bank Interest Rate	(83,503)	(91,419)	(100,086)	(109,574)	(119,961)	(131,334)	(143,784)	(157,415)	(172,338)	(188,675)
Loans	880,834	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,468,057	0	0	0	0	0	0	0	0	0
Net (Increase/Decrease) in Cash	1,086,492	427,687	440,798	454,379	468,455	483,050	498,191	513,907	530,229	547,189
Cashflows at the Beginning of Period	0	1,086,492	1,514,179	1,954,977	2,409,356	2,877,811	3,360,861	3,859,052	4,372,959	4,903,187
Cashflows at the End of Period	1,086,492	1,514,179	1,954,977	2,409,356	2,877,811	3,360,861	3,859,052	4,372,959	4,903,187	5,450,377

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 37 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	1,086,492	1,514,179	1,954,977	2,409,356	2,877,811	3,360,861	3,859,052	4,372,959	4,903,187	5,450,377
Inventory	173,938	179,156	184,531	190,067	195,769	201,642	207,691	213,922	220,340	226,950
Total current Assets	**1,260,430**	**1,693,335**	**2,139,508**	**2,599,423**	**3,073,580**	**3,562,503**	**4,066,743**	**4,586,881**	**5,123,527**	**5,677,327**
Non Current Assets										
Fixed Assets (net)	652,300	595,793	538,336	479,908	420,490	360,061	298,599	236,082	172,489	107,797
Total Non Current Assets	**652,300**	**595,793**	**538,336**	**479,908**	**420,490**	**360,061**	**298,599**	**236,082**	**172,489**	**107,797**
Total Assets	**1,912,730**	**2,289,128**	**2,677,844**	**3,079,331**	**3,494,070**	**3,922,563**	**4,365,342**	**4,822,963**	**5,296,016**	**5,785,123**
Liabilities										
Paybles	84,828	87,373	89,994	92,694	95,475	98,339	101,289	104,328	107,458	110,682
Remaining amount of Loan	(91,419)	(100,086)	(109,574)	(119,961)	(131,334)	(143,784)	(157,415)	(172,338)	(188,675)	0
Total current Liabilities	**(6,591)**	**(12,712)**	**(19,579)**	**(27,267)**	**(35,859)**	**(45,445)**	**(56,125)**	**(68,010)**	**(81,217)**	**110,682**
Non Current Liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Long Terms Loans	1,055,756	1,155,842	1,265,416	1,385,377	1,516,711	1,660,495	1,817,910	1,990,248	2,178,923	2,178,923
Total Long Term Liabilities	**1,055,756**	**1,155,842**	**1,265,416**	**1,385,377**	**1,516,711**	**1,660,495**	**1,817,910**	**1,990,248**	**2,178,923**	**2,178,923**
Total Liabilities	**1,049,165**	**1,143,129**	**1,245,836**	**1,358,110**	**1,480,852**	**1,615,050**	**1,761,784**	**1,922,238**	**2,097,706**	**2,289,605**
Owners Equity										
Shareholders Contributions	587,223	587,223	587,223	587,223	587,223	587,223	587,223	587,223	587,223	587,223
Statutory Reseve	27,634	55,878	84,478	113,400	142,600	172,029	201,633	231,350	261,109	290,830
Retained Profits	248,708	502,899	760,306	1,020,599	1,283,396	1,548,262	1,814,701	2,082,152	2,349,979	2,617,466
Total Equity	863,565	1,145,999	1,432,007	1,721,221	2,013,218	2,307,513	2,603,557	2,900,725	3,198,310	3,495,518
Total liabilities and equity	**1,912,730**	**2,289,128**	**2,677,844**	**3,079,331**	**3,494,070**	**3,922,563**	**4,365,342**	**4,822,963**	**5,296,016**	**5,785,123**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 38 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(1,468,057)	340,58	427,68	440,79	454,37	468,45	483,05	498,19	513,90	530,22	547,18	8,426,874
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value	**(1,468,057)**	**250,70**	**284,25**	**264,52**	**246,19**	**229,18**	**213,37**	**198,69**	**185,06**	**172,40**	**160,64**	**2,473,973**

Table below (38) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table39 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free	(1,468,057)	340,585	427,68	440,79	454,379	468,455	483,050	498,191	513,907	530,229	547,189
Project Value	1,468,057	1,127,47	699,78	258,98	)	)	)	)	)	)	3,236,414)
Payback Period	3	1	1	1	0	0	0	0	0	0	0

Table 40 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	3,210,953
Payback Period	3
Internal Rate of Return	%35.00

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 41 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%35.00	3	10.75%
Revenues declined by 10%	%33.07	3	10.75%
Operating Expenses Increased by 10%	%20.03	6	10.75%

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 3,210,953 Jordan Dinars considering that the project provides 13 Job Opportunitites for the governorate residents.